

EDUCATION

- Bachelor in Engineering** | *Electronics and Communication* | **GPA: 3.22** Nov. 2019 – April 2024
Tribhuwan University, Institute Of Engineering, Pulchowk Campus, Pulchowk, Nepal
Major subjects: Basic-Advanced Electronics, Microprocessor, C++ , Embedded Systems, Control Systems, Filter Design, Digital Signal Analysis and Processing,Computer Architecture, Electromagnetics, Data Structure, Digital Logics, Communication Systems,Telecommunication Systems, Antenna, RF microwave and AI etc.

Major Interests: VLSI Design, Hardware Accelerators In-Memory Computing, Secure Designs, Microelectronics Systems, CAD Tools.

SKILLS

Technical Skills:	Personal Skills	English Language Proficiency:
Programming: Programming fundamental and OOP Concepts, C++ , C, JAVA, Python, SystemVerilog(Design+Validation), RUST Hardware: RISC-V Architecture based processor design and testing(Learning), 8085/8086 MPU, 8051/52 microcontroller, Assembly, KICAD designing, PCB fabrication, drilling, soldering, Arduino, MCU-peripheral interfacing. Tools and libs:Git, LaTeX, Proteus, Modelsim ML tools and Libs:Numpy, Pandas, PyTorch and Sklearn(Learning) Others: Involved in Computer Vision, Machine Learning, Deep Learning and AI.	- Teamwork and Collaboration - Highly Organized and Disciplined - Critical Thinking and Problem Solving - Adaptability with working environment and Open Minded - Effective Communications	*Level: C1 *IELTS: 7

PUBLICATIONS

- S. K. Khatri, D. Parajuli, P. M. Mane, B. Chaulagain, and S. Thapa, “**Fall Detection System for Elderly People Using LSTM**”, J. Eng. Iss. Solut., vol. 5, no. 1, pp. 210–220, Apr. 2026.
This study addresses the consequences and effects of falls on elderly people and provides a low-powered wearable hardware solution embedded with an LSTM model for detecting fall events to enhance elderly safety.

PROJECTS

- Design and Verification of a Fixed-Point Attention Accelerator** | *SystemVerilog/Python* 2026
Implemented a accelerator for the self-attention mechanism using SystemVerilog and fixed-point arithmetic.
- Internet of Medical Things and Streaming** | *C++/Python* 2025
Contributed in sensors and MQTT protocol configuration; Developed Interactive Dashboards.
- RISC-V-Complete-Multicycle-Processor Design** | *SystemVerilog* 2024
Modeled RISC-V complete multicycle processor and designed using SystemVerilog.
- Design of Star Tracker Simulator for the Attitude Determination of Spacecrafts** 2024
Developed of Image Simulator Software; inegrated hardware components with the PI processor.
- Fall Detection System** | *Python, Hardware System* 2023
Designed the circuitry for the wearable device in KICAD and fabricated it in PCB

- **Dijkstra's Algorithm Visualizer** | C++ 2022
Implemented the concept of Data Structure like Nodes, linked list, Stack, Tree, Map etc in this program.
- **OCR Using OpenCV and Google's Tesseract API** | C++ 2021
A small tool for optical character recognition; Contributed in making environment setup, and application development; Implemented OpenCV in C++.

WORK EXPERIENCE

- **Part time Lecturer** Apr 2025–Present
Lalitpur Engineering College, IOE, Lalitpur, Nepal
 - Lectured undergraduate Electronics Engineering courses in Microprocessors, Computer Organization and Architecture and Embedded Systems.
 - Conducted practical laboratory sessions involving digital circuit design, microcontroller programming and hardware interfacing.
 - Guided students in practical implementations, debugging, and project-based learning activities.
- **Software Engineer in Test** June 2024–Present
Cedar Gate Technologies, Nepal site office, Sanepa Lalitpur, Nepal
 - Worked on SDLC processes, Cloud Computing and developed enterprise product knowledge base for GenAI applications by defining business rules and designing LLM skills.
 - Enhanced interpersonal skills, documentations skills and learned about professional ethics.
- **Teaching Assistant** Aug 2024-Nov 2024
IOE, Pulchowk Campus
 - Mentored students in core Python programming concepts for the Samsung Innovation Campus (SIC), focusing on problem-solving, code optimization, and clean coding practices
 - Facilitated hands-on learning through supervised projects and coding sessions, while assisting instructors in content delivery and interactive session management to ensure effective learning outcomes.
- **Undergraduate Research Fellow** | *IOE Pulchowk Campus* Nov 2022-May 2024
 - Conducted undergraduate research on reliable fall detection systems using inertial sensor data and machine learning techniques.
 - Investigated LSTM-based approaches for detecting and classifying human falls from time-series IMU data.
 - Worked on sensor data processing, algorithm development, and experimental validation for research prototypes.
 - Contributed to the design of image simulator for star tracker for attitude determination applications.
 - Presented research findings and contributed to the technical documentation and development of research projects.
- **Hardware Coordinator** | *Robotics Club, IOE Pulchowk Campus* 2023-2024
 - Coordinated hardware events on Campus; Actively participated in ABU Robocon Competition.